

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**

**Product Identifier**

Perihalan Produk:

Product Description:

Cat No. :

Puck's F Balanced Salt Solution (1X), with calcium, with mag

Puck's F Balanced Salt Solution (1X), with calcium, with mag

J67665

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use

Laboratory chemicals.

Uses advised against

No Information available

**Company**

Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
Selangor Darul Ehsan, Malaysia.  
Main line: +60 3-5525 7888

**Supplier**

E-mail address

Enquiry.my@thermofisher.com

Emergency Telephone Number

Tel: +03-5525 7888

CHEMTREC Malaysia **1-800-815-308** (Malay)

CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**

**Classification of the substance or mixture**

**Label Elements**

**Hazard Statements**

**Other Hazards**

This product does not contain any known or suspected endocrine disruptors

**SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**

# SAFETY DATA SHEET

Puck's F Balanced Salt Solution (1X), with calcium, with mag

Revision Date 31-Mar-2025

| Component                      | CAS No     | Weight % |
|--------------------------------|------------|----------|
| Water                          | 7732-18-5  | 98.65    |
| Sodium chloride                | 7647-14-5  | 0.8      |
| Potassium chloride             | 7447-40-7  | 0.395    |
| Glucose                        | 50-99-7    | 0.11     |
| Sodium phosphate dibasic       | 7558-79-4  | 0.016    |
| Dihydrogen potassium phosphate | 7778-77-0  | 0.015    |
| Magnesium sulfate              | 7487-88-9  | 0.008    |
| Phenol red                     | 143-74-8   | 0.004    |
| Calcium chloride               | 10043-52-4 | 0.001    |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

**Eye Contact** Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact** Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

**Ingestion** Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

**Inhalation** Remove to fresh air. Get medical attention immediately if symptoms occur.

**Self-Protection of the First Aider** No special precautions required.

### Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### Indication of any immediate medical attention and special treatment needed

**Notes to Physician** Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Not combustible.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

None reasonably foreseeable.

### **Hazardous Combustion Products**

Sulfur oxides, Hydrogen chloride, Oxides of phosphorus, Calcium oxides, Potassium oxides, Sodium oxides.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

# SAFETY DATA SHEET

Puck's F Balanced Salt Solution (1X), with calcium, with mag

Revision Date 31-Mar-2025

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required.

### Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation.

### Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed in a dry and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

### Exposure Controls

#### Engineering Measures

None under normal use conditions.

### Personal protective equipment

#### **Eye Protection**

Wear safety glasses with side shields (or goggles)

#### **Hand Protection**

Protective gloves

#### **Skin and body protection**

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

# SAFETY DATA SHEET

Puck's F Balanced Salt Solution (1X), with calcium, with mag

Revision Date 31-Mar-2025

|                                 |                                                               |
|---------------------------------|---------------------------------------------------------------|
| <b>Respiratory Protection</b>   | No protective equipment is needed under normal use conditions |
| <b>Recommended Filter type:</b> | Particle filter                                               |

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice

**Environmental exposure controls** No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                     |                          |                                          |
|-------------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>                   | Red                      |                                          |
| <b>Physical State</b>               | Liquid                   |                                          |
| <b>Odor</b>                         | No information available |                                          |
| <b>Odor Threshold</b>               | No data available        |                                          |
| <b>pH</b>                           | No information available |                                          |
| <b>Melting Point/Range</b>          | No data available        |                                          |
| <b>Softening Point</b>              | No data available        |                                          |
| <b>Boiling Point/Range</b>          | No information available |                                          |
| <b>Flash Point</b>                  | No information available | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>             | No data available        |                                          |
| <b>Flammability (solid,gas)</b>     | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>             | No data available        |                                          |
| <b>Vapor Pressure</b>               | 23 hPa @ 20 °C           |                                          |
| <b>Vapor Density</b>                | No data available        | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>   | No data available        |                                          |
| <b>Bulk Density</b>                 | Not applicable           | Liquid                                   |
| <b>Water Solubility</b>             | Miscible                 |                                          |
| <b>Solubility in other solvents</b> | No information available |                                          |

### **Partition Coefficient (n-octanol/water)**

|                                  |                          |
|----------------------------------|--------------------------|
| <b>Autoignition Temperature</b>  | No data available        |
| <b>Decomposition Temperature</b> | No data available        |
| <b>Viscosity</b>                 | No data available        |
| <b>Explosive Properties</b>      | No information available |
| <b>Oxidizing Properties</b>      | No information available |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

# SAFETY DATA SHEET

Puck's F Balanced Salt Solution (1X), with calcium, with mag

Revision Date 31-Mar-2025

## Chemical Stability

Stable under normal conditions.

## Possibility of Hazardous Reactions

**Hazardous Polymerization** No information available.  
**Hazardous Reactions** None under normal processing.

## Conditions to Avoid

None known.

## Incompatible Materials

None known.

## Hazardous Decomposition Products

Sulfur oxides. Hydrogen chloride. Oxides of phosphorus. Calcium oxides. Potassium oxides. Sodium oxides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

**Oral** Based on available data, the classification criteria are not met  
**Dermal** Based on available data, the classification criteria are not met  
**Inhalation** Based on available data, the classification criteria are not met

### Toxicology data for the components

| Component                      | LD50 Oral                 | LD50 Dermal                   | LC50 Inhalation              |
|--------------------------------|---------------------------|-------------------------------|------------------------------|
| Water                          | -                         | -                             | -                            |
| Sodium chloride                | LD50 = 3550 mg/kg ( Rat ) | LD50 > 10000 mg/kg ( Rabbit ) | LC50 > 42 mg/L ( Rat ) 1 h   |
| Potassium chloride             | LD50 = 2600 mg/kg ( Rat ) | -                             | -                            |
| Glucose                        | 25.8 g/kg ( Rat )         | -                             | -                            |
| Sodium phosphate dibasic       | LD50 = 17 g/kg ( Rat )    | -                             | -                            |
| Dihydrogen potassium phosphate | LD50 = 3200 mg/kg ( Rat ) | LD50 > 4640 mg/kg ( Rabbit )  | LC50 > 0.83 mg/L ( Rat ) 4 h |
| Calcium chloride               | 2301 mg/kg ( Rat )        | LD50 > 5000 mg/kg ( Rabbit )  | -                            |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;  
**Respiratory** No data available

# SAFETY DATA SHEET

Puck's F Balanced Salt Solution (1X), with calcium, with mag

Revision Date 31-Mar-2025

|                                                   |                                                                                                                                     |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin</b>                                       | No data available                                                                                                                   |
| <b>(e) germ cell mutagenicity;</b>                | No data available                                                                                                                   |
| <b>(f) carcinogenicity;</b>                       | No data available<br>There are no known carcinogenic chemicals in this product                                                      |
| <b>(g) reproductive toxicity;</b>                 | No data available                                                                                                                   |
| <b>(h) STOT-single exposure;</b>                  | No data available                                                                                                                   |
| <b>(i) STOT-repeated exposure;</b>                | No data available                                                                                                                   |
| <b>Target Organs</b>                              | No information available.                                                                                                           |
| <b>(j) aspiration hazard;</b>                     | No data available                                                                                                                   |
| <b>Symptoms / effects, both acute and delayed</b> | No information available.                                                                                                           |
| <b>Endocrine Disrupting Properties</b>            | Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors. |

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects** Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

| Component          | Freshwater Fish                                                                                    | Water Flea                                           | Freshwater Algae                                 | Microtox                                            |
|--------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------|
| Sodium chloride    | Pimephals prome:<br>LC50: 7650 mg/L/96h                                                            | EC50: 1000 mg/L/48h                                  |                                                  |                                                     |
| Potassium chloride | Lepomis macrochirus:<br>LC50: 1060 mg/L /96h<br>Pimephales promelas:<br>LC50: 750 - 1020 mg/L /96h | EC50: 825 mg/L/48h                                   | EC50: 2500 mg/L/72h                              |                                                     |
| Magnesium sulfate  | LC50: 2610 - 3080 mg/L, 96h static (Pimephales promelas)                                           | EC50: 266.4 - 417.3 mg/L, 48h Static (Daphnia magna) | EC50: = 2700 mg/L, 72h (Desmodesmus subspicatus) | = 84000 mg/L EC50 Photobacterium phosphoreum 30 min |
| Calcium chloride   | Lepomis macrochirus:<br>LC50: 10650 mg/L/96h                                                       | EC50: 52 mg/L/48h                                    |                                                  |                                                     |

### Persistence and degradability

#### **Persistence**

Miscible with water, Persistence is unlikely, based on information available.

### Bioaccumulative potential

Bioaccumulation is unlikely

### Mobility in soil

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

# SAFETY DATA SHEET

Puck's F Balanced Salt Solution (1X), with calcium, with mag

Revision Date 31-Mar-2025

Other adverse effects No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods

**Waste from Residues/Unused Products**

Dispose of in accordance with local regulations

**Contaminated Packaging**

Empty containers should be taken to an approved waste handling site for recycling or disposal

## SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

Not regulated

Road and Rail Transport

Not regulated

IATA

Not regulated

**Special Precautions for User**

No special precautions required

## SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

**International Inventories**

X = listed

| Component                      | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|--------------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Water                          | 231-791-2 | X    | X   | X     | X    |      | X     | X    | KE-35400 |
| Sodium chloride                | 231-598-3 | X    | X   | X     | X    | X    | X     | X    | KE-31387 |
| Potassium chloride             | 231-211-8 | X    | X   | X     | X    | X    | X     | X    | KE-29086 |
| Glucose                        | 200-075-1 | X    | X   | X     | X    | X    | X     | X    | KE-17727 |
| Sodium phosphate dibasic       | 231-448-7 | X    | X   | X     | X    | X    | X     | X    | KE-12344 |
| Dihydrogen potassium phosphate | 231-913-4 | X    | X   | X     | X    | X    | X     | X    | KE-28622 |
| Magnesium sulfate              | 231-298-2 | X    | X   | X     | X    | X    | X     | X    | KE-22752 |
| Phenol red                     | 205-609-7 | X    | X   | X     | X    | X    | X     | X    | KE-02748 |
| Calcium chloride               | 233-140-8 | X    | X   | X     | X    | X    | X     | X    | KE-04496 |

National Regulations

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

# SAFETY DATA SHEET

Puck's F Balanced Salt Solution (1X), with calcium, with mag

Revision Date 31-Mar-2025

## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Prepared By**

**Revision Date**

**Revision Summary**

Health, Safety and Environmental Department

31-Mar-2025

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**